

Homework 8. PCR and PLS

Stat 427/627, Statistical Machine Learning

See due date on Canvas

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1 College Applications (p.287, 16 pts) 1

*Total: 16 pts.** This assignment covers PCR and PLS. It is a continuation of the College Application analysis in HW 2 and HW 7.

0.1 Instructions:

- You only need to work on sub-questions (e), (f), (g), (h), (i) in this assignment. You have answered sub-questions (a) ~ (d) in previous homework.
- Numerical results in this assignment may vary due to randomization.

Question	a	b	c	d	e	f	g	h	i	Total
427	-	-	-	-	4	4	4	4		16
627	-	-	-	-	4	4	3	3	2	16

1 College Applications (p.287, 16 pts)

Predict the number of applications received based on the other variables in the `College` data set in `{ISLR2}`.

Use the following code to split the data set randomly. Use the subset `college.data` for this assignment. Set aside `holdout.data` data for the next assignment.

```
library(ISLR2)
data("College")

my.college <- College[-484, ] # remove an extreme case.
#my.college <- College[College$Apps <=16000, ] # remove several extreme case.
train.pct <- 0.78
set.seed(2025)
Z <- sample(nrow(my.college), floor(train.pct*nrow(my.college)))
college.data <- my.college[Z, ]
holdout.data <- my.college[-Z, ]
```

Recall that we fitted the (training) data in Homework 2.

- We fit the **full** model with all 17 predictors. That is:

```
college.lmF <- lm(Apps ~ ., data = college.data)
college.lmF
```

- Using stepwise selection, AIC is the smallest with 11 predictors (12 parameters):
 $\text{Apps} \sim \text{Private} + \text{Accept} + \text{Top10perc} + \text{Top25perc} + \text{F.Undergrad} + \text{Outstate} + \text{Room.Board} + \text{PhD} + \text{perc.alumni} + \text{Expend} + \text{Grad.Rate},$
- Using the best-subset algorithm. BIC is the smallest with 8 predictors (9 parameters):
 $\text{Apps} \sim \text{Private} + \text{Accept} + \text{Top10perc} + \text{Top25perc} + \text{Outstate} + \text{Room.Board} + \text{PhD} + \text{Expend}$

Continue working on the training data set (`college.data` from above) and the above models.

- (4 pt) Compute the Variance Inflation Factor (VIF) and comment on the severity of the collinearity of the data. Why is “collinearity” a concern, even if the model is correct?
- (4 pts) Evaluate prediction accuracy of your selected models based on AIC and BIC respectively. Estimate the prediction mean squared error by 10-fold cross-validation. Recall that `glm( ..., family=gaussian)` fits linear regression and its outcome can be used in `cv.glm()` (boot package) for cross-validation.
- (4 pts) Consider the **full** model with all 17 predictors (reminder: use data frame `college.data` you recreated at the beginning). Use functions in package `glmnet` to fit a ridge regression.
 - Select λ chosen by (default 10-fold) cross-validation.
 - Plot the results of the cross-validation.
 - Report the estimated MSE of the model based on your selected λ .
- (4 pts) Consider the **full** model with all 17 predictors (reminder: use data frame `college.data` you recreated at the beginning). Use functions in package `glmnet` to fit a LASSO regression.
 - Select λ chosen by (default 10-fold) cross-validation.
 - Plot the results of the cross-validation.
 - Report the estimated MSE of the model based on your selected λ .
- (4 pts) Fit a PCR model on `college.data`, with M (the number of principal components) chosen by cross validation. Prepare a validation plot. Report the estimated test error (MSE), along with the value of M selected by cross-validation.
- (4 pts) Fit a PLS (partial least squares) model on `college.data`, with M (the number of principal components) chosen by cross validation. Prepare a validation plot. Report the estimated test error (MSE), along with the value of M selected by cross-validation.
- (3 pts) Summarize and comment on the results obtained from the following models. Recommend a model, and justify your choice.

Method	Number of predictors	Estimated prediction MSE
Least Squares 1: model with the smallest AIC (See HW 7)		
Least Squares 2: model with the smallest BIC (See HW 7)		
Ridge Regression (lambda.min) (See HW 7)		
Lasso (lambda.min) (See HW 7)		
Lasso (lambda.1se) (See HW 7)		
PCR		
PLS		

- (h) (3 pts) Apply the above models to the hold-out data `holdout.data` that we created at the beginning. Which model wins this contest in terms of prediction accuracy? (This should be the first time you use observations in `holdout.data` data frame.)
- (i) (2 pts) **Stat-627** Compare your estimated prediction MSE from the training data `college.data` (part i) and the resulting MSE from the `holdout.data` (part j). Is there anything “surprising” that worth investigation? If yes, what are the possible causes? (Note. It is not surprising to see a tuned “best” model not to perform the best on the testing data.)

—— This is the end of HW 8. ——